

# Python Operators

## Day 2

- Arithmetic operators
- Assignment operators
- Comparison operators & Logical operators
- Identity operators
- Membership operators & Output
- Simple Output Formatting

### 1. Arithmetic Operators:

Arithmetic operators are used for mathematical operations. The basic arithmetic operators in Python are:

-  (Addition): Adds two operands.
-  (Subtraction): Subtracts the right operand from the left operand.
-  (Multiplication): Multiplies two operands.
-  (Division): Divides the left operand by the right operand.
-  (Modulus): Returns the remainder of the division of the left operand by the right operand.
-  (Floor Division): Returns the quotient of the division, discarding the remainder.
-  (Exponentiation): Raises the left operand to the power of the right operand.

### Example:

```
a = 10
b = 3
```

```
addition = a + b
subtraction = a - b
multiplication = a * b
division = a / b
remainder = a % b
floor_division = a // b
exponentiation = a ** b

print(addition, subtraction, multiplication, division, remainder)
```

## 2. Assignment Operators:

Assignment operators are used to assign values to variables. The basic assignment operator is `=`. There are also compound assignment operators like `+=`, `-=`, `*=`, etc., which combine an operation with assignment.

### Examples:

```
x = 5
x += 3 # equivalent to x = x + 3
y = 10
y *= 2 # equivalent to y = y * 2

print(x, y)
```

## 3. Comparison Operators:

Comparison operators are used to compare values. They return a Boolean result (`True` or `False`).

- `==` (Equal to)
- `!=` (Not equal to)
- `<` (Less than)
- `>` (Greater than)
- `<=` (Less than or equal to)

- `>=` (Greater than or equal to)

#### Examples:

```
a = 5
b = 10

print(a == b)  # False
print(a != b)  # True
print(a < b)   # True
print(a > b)   # False
```

## 4. Logical Operators:

Logical operators are used to combine multiple conditions. The basic logical operators are:

- `and` (Logical AND)
- `or` (Logical OR)
- `not` (Logical NOT)

#### Examples:

```
x = True
y = False

print(x and y)  # False
print(x or y)   # True
print(not x)    # False
```

## 5. Identity Operators:

Identity operators are used to check if two variables refer to the same object.

- `is` : Returns `True` if both variables are the same object.
- `is not` : Returns `True` if both variables are not the same object.

#### Examples:

```
a = [1, 2, 3]
b = a
c = [1, 2, 3]

print(a is b)      # True
print(a is not c)  # True
```

## 6. Membership Operators:

Membership operators are used to test whether a value is a member of a sequence (e.g., a list, tuple, or string).

- `in`: Returns `True` if a value is found in the sequence.
- `not in`: Returns `True` if a value is not found in the sequence.

### Examples:

```
list_example = [1, 2, 3, 4, 5]

print(3 in list_example)    # True
print(6 not in list_example) # True
```

## 7. Output Formatting:(F-strings)

Output formatting allows you to create dynamic strings by inserting values into placeholders.

F-strings, introduced in Python 3.6, are a concise and readable way

Easily insert variables into a string.

You can perform calculations or call functions directly within the string.

### Example:

```
name = "John"
age = 25
```

```
# Using f-string for formatting  
print(f"My name is {name} and I am {age} years old.")
```